

Isotopes and Average Atomic Mass

Answer key with worked solutions

A. Understanding

1. What the atomic mass represents.

The **weighted average** mass of the element's naturally occurring isotopes, based on their abundances.

2. Which isotope is more abundant?

The **20.0 u** isotope: the average (20.2 u) is much closer to 20.0 than to 22.0, so the lighter isotope must be far more common.

B. Calculating average atomic mass

3. Boron.

$$(0.199 \times 10.013) + (0.801 \times 11.009) = \mathbf{10.81 \text{ u}}$$

4. Rubidium.

$$(0.7217 \times 84.912) + (0.2783 \times 86.909) = \mathbf{85.47 \text{ u}}$$

5. Silver.

$$(0.5184 \times 106.905) + (0.4816 \times 108.905) = \mathbf{107.87 \text{ u}}$$

6. Silicon.

$$(0.9223 \times 27.977) + (0.0468 \times 28.976) + (0.0309 \times 29.974) = \mathbf{28.09 \text{ u}}$$

C. Challenge

7. Abundances of copper isotopes.

Let x = fraction of copper-63: $62.930x + 64.928(1 - x) = 63.546$

$64.928 - 1.998x = 63.546$, so $x = \frac{1.382}{1.998} = 0.692$

Copper-63: 69.2%, copper-65: 30.8%